

# Proposal

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## 1. Necessity of the R&D Project

- Quantum network systems theoretically provide **unconditional security** through quantum key distribution (QKD) based on the principles of quantum mechanics, but in practice, **various vulnerabilities may exist**.
  - However, **security research on quantum systems** has not yet been actively conducted. In particular, studies on **security threats and the attack surface** remain insufficient.
  - Furthermore, security research on quantum systems requires a **broad understanding** of various domains including **communications, networks, and software**, making **interdisciplinary research essential**.
  - From the **network perspective**, it is unrealistic for a QKD system to operate independently. It must be **closely integrated** with widely used conventional network security protocols such as **TLS and IPSec**. Thus, **clear interface design and management** between quantum and classical protocols are essential.
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**[Figure]** Conceptual diagram of quantum systems and vulnerabilities using quantum communication

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## 2. Objectives and Details of the R&D Project

### 2-1. Objectives of the R&D Project

(Network – Prof. Min Suk Kang)

- Design and implement an **integrated protocol** that provides **practical security** by **clearly and efficiently designing** the interface between QKD technology and existing security protocols (e.g., TLS).
  - Develop a **protocol structure** that **minimizes implementation-related security vulnerabilities**.
  - **Evaluate** the performance and operational efficiency of the proposed protocol in **real network environments** to verify **practical applicability**.
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### 2-2. Year 1

(Network)

**Detailed Research Objective 1: Analysis of existing quantum-classical integration protocols** (compared with [1], [2])

- Conduct an in-depth analysis of **QKD-KEM** [1], a recently proposed QKD and security protocol integration technique, to identify its **strengths, weaknesses, and feasibility**.
- QKD-KEM [1] uses **OpenSSL's Provider framework** to integrate QKD keys directly into the **TLS handshake phase**. While it increases deployment feasibility by minimizing infrastructure changes, it **fails to clearly address key management and operational complexity**.
- This study builds on that approach but **further designs a common interface** to enable expansion beyond TLS to other protocols such as **IPSec**, thus creating a **generalized integration framework**.
- In addition, by analyzing [2], which summarizes **recent trends in QKD-based secure communication protocols**, the study examines the **practical issues** these **theoretical**

**protocols** may face in real-world networks.

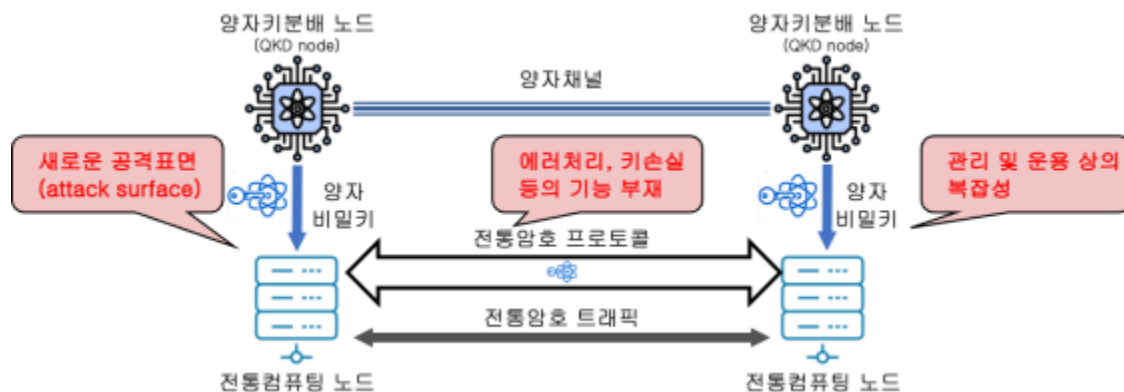
[1] Javier Blanco-Romero et al., "QKD-KEM: Hybrid QKD Integration into TLS with OpenSSL Providers," *arXiv preprint* arXiv:2503.07196 (2025).

[2] Agulleiro et al., "Quantum Cryptography Protocols Ensuring Secure Communication," *ITM Web of Conferences* 50, 05009 (2025).

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## Detailed Research Objective 2: Interface design between quantum and classical protocols

- Clearly define an **interface structure** to insert QKD-generated keys into existing TLS key exchange mechanisms, **minimizing complexity** in management and operations.
- Design and define a **common interface** applicable not only to TLS but also to other protocols such as **IPSec**, enabling **multi-protocol integration**.
- Include a **key management framework** that explicitly defines the **entire lifecycle** of quantum keys — from generation to destruction — to enhance security from the design phase.
- Clearly specify procedures for **error handling**, **key loss**, and **renegotiation** during integration with classical protocols to ensure **stability**.
- Design **recovery procedures** for realistic failure scenarios (e.g., **temporary quantum channel disruption**, protocol mismatch), to build a **resilient and robust** protocol from the ground up.



**[Figure]** Conceptual diagram of quantum-classical interface design

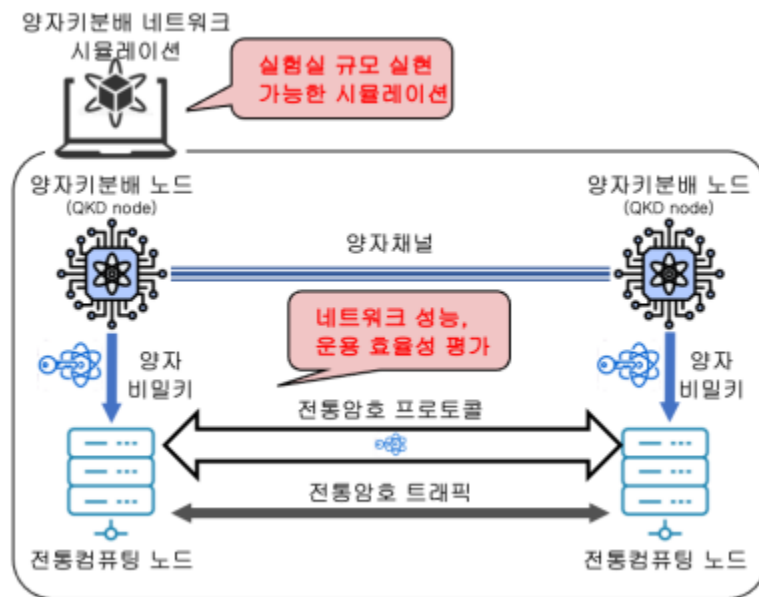
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## 2-3. Year 2

### (Network)

**Detailed Research Content: Implementation of integrated protocol prototype and analysis of operational efficiency**

- Implement the **QKD-TLS integration protocol** designed in Year 1 as a **prototype**, test it in real network environments, and evaluate experimentally.
  - Measure network **performance metrics** (e.g., latency, throughput, key exchange speed) to verify **efficiency** and assess **real-world applicability**.
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**[Figure]** Conceptual diagram of integrated protocol

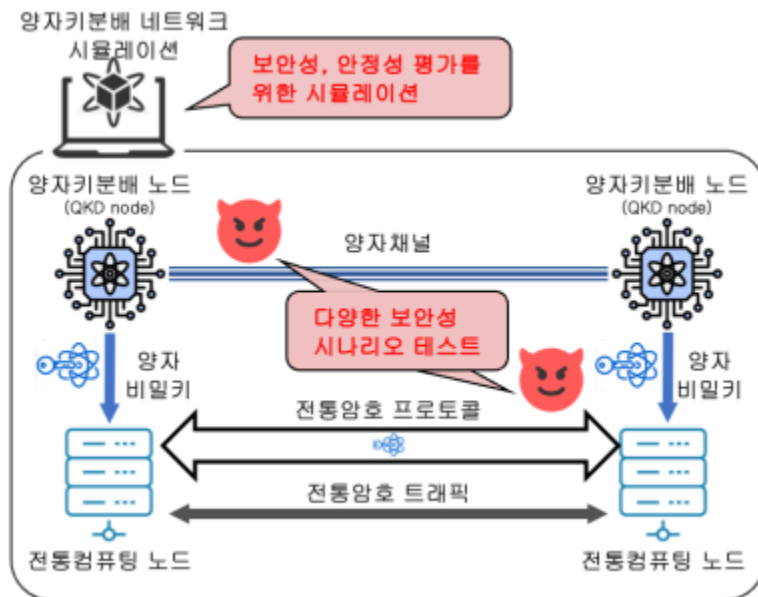
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## 2-4. Year 3

### (Network)

**Detailed Research Content: Security and stability evaluation of the integrated protocol and final demonstration**

- Evaluate the **security** of the integration with classical protocols via **simulations and experiments** in various scenarios.
- Implement and assess **methods to reduce complexity** in key management and transmission across the quantum-classical interface.
- **Submit results to international conferences and journals**, and **release the final implementation** publicly.



**[Figure]** Conceptual diagram of final demonstration of the integrated protocol